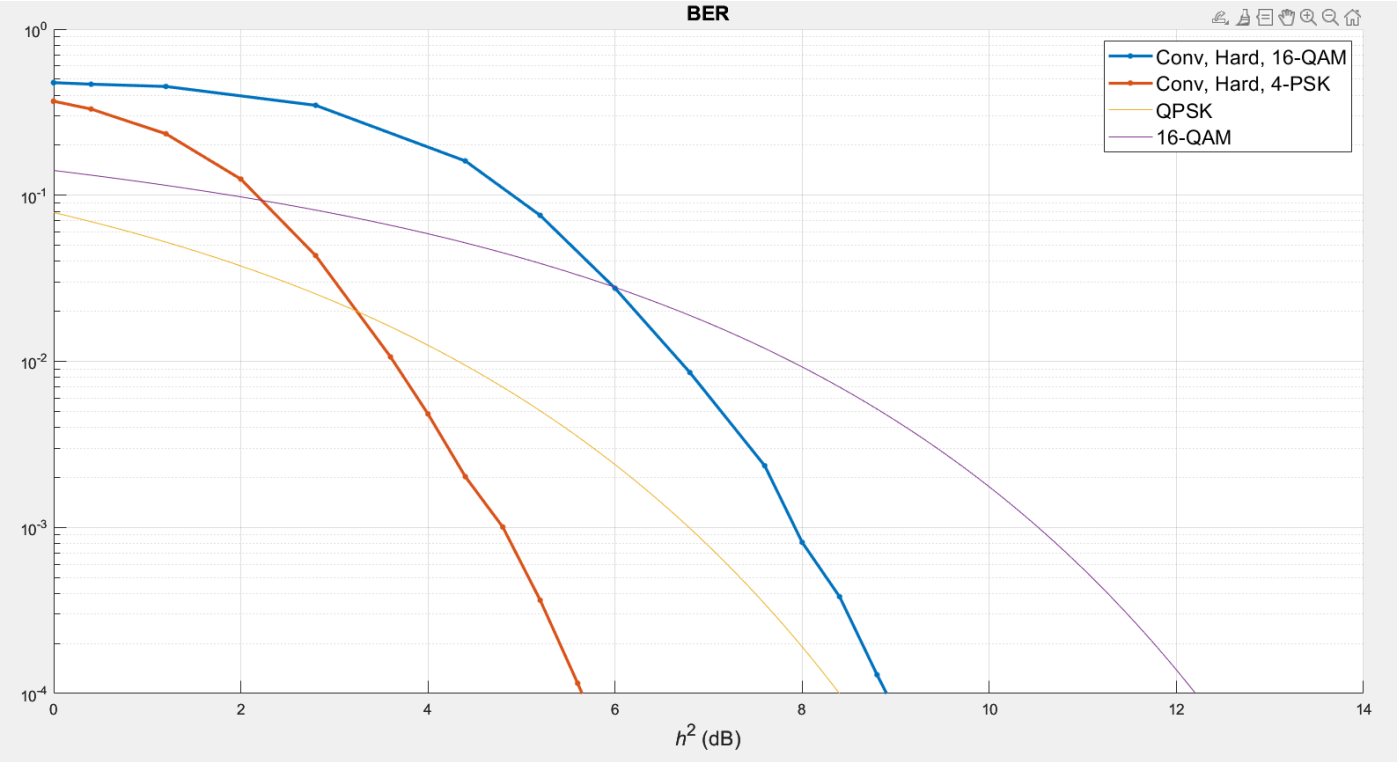
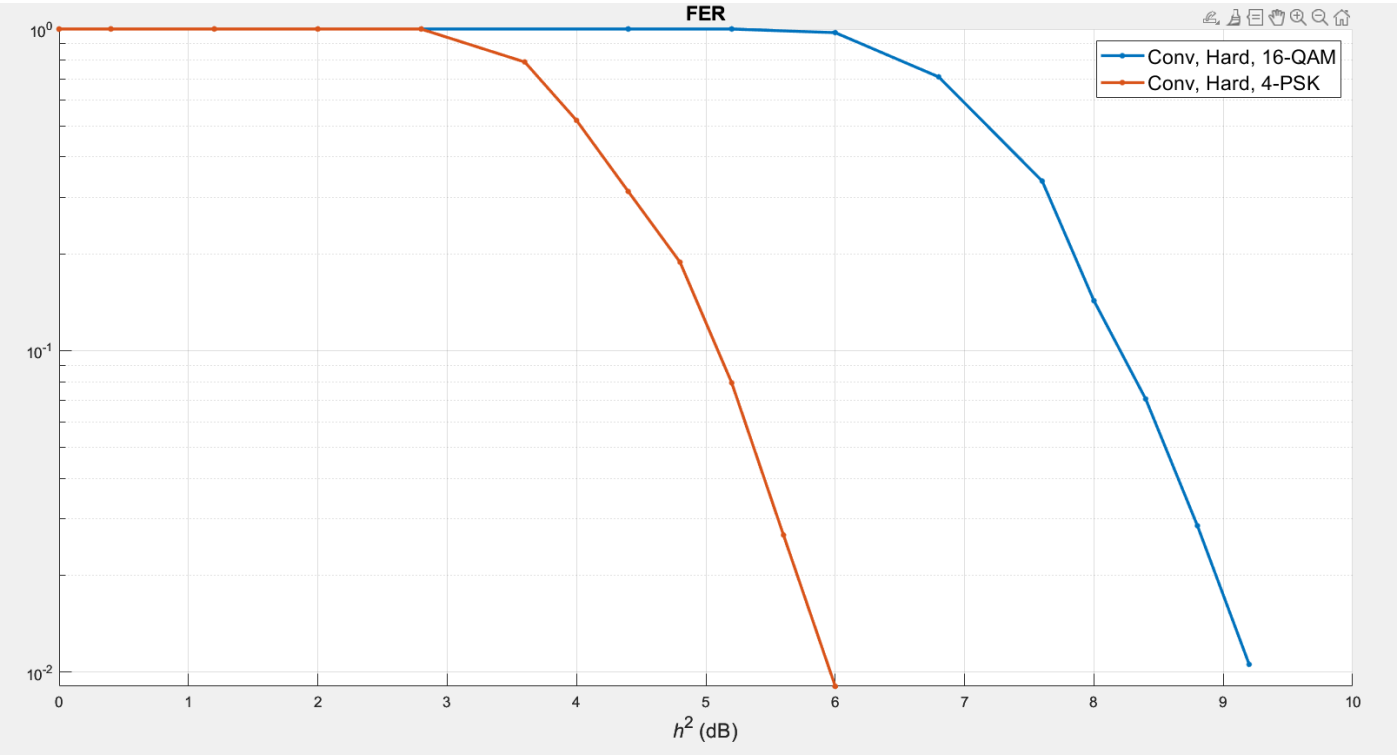
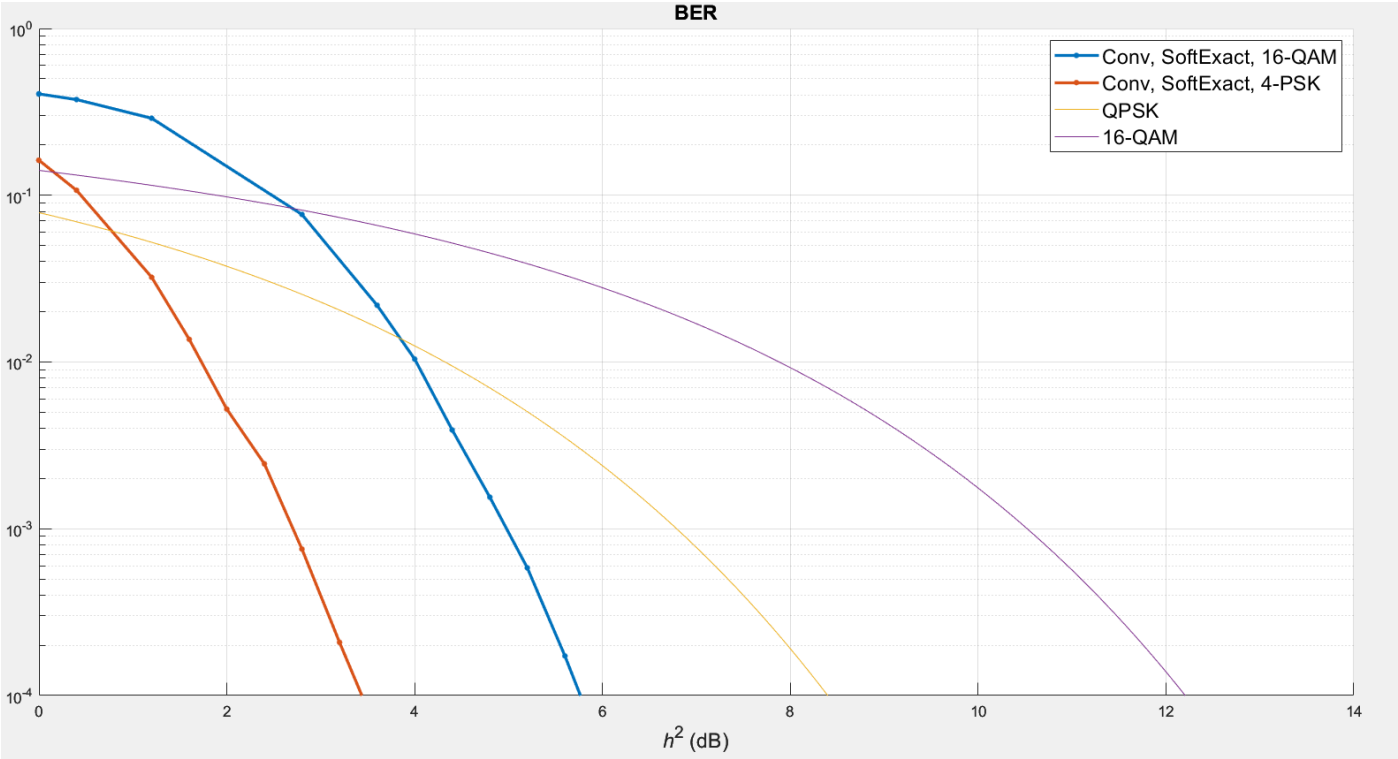
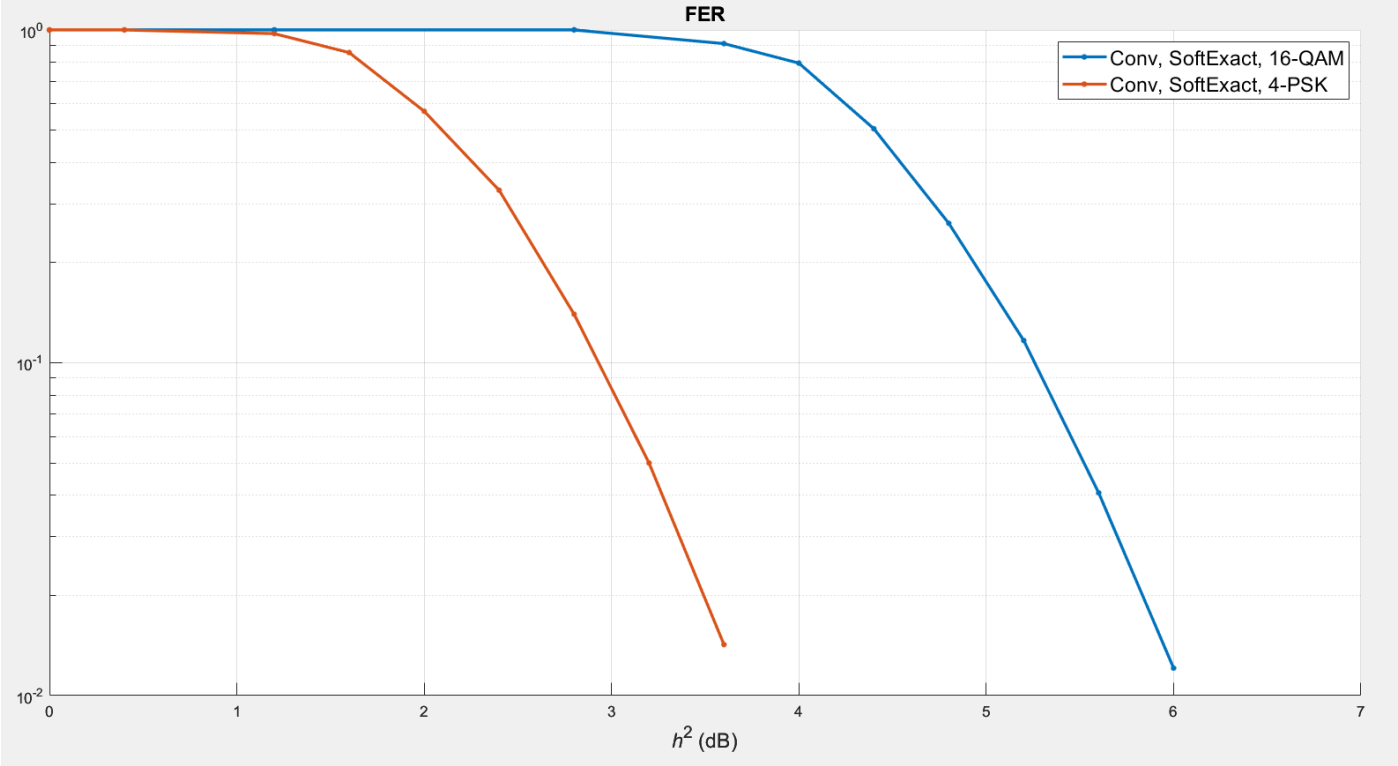
1.1) Жесткие Решения Демодулятора, сверточный код для созвездий QAM-16 и PSK-4.



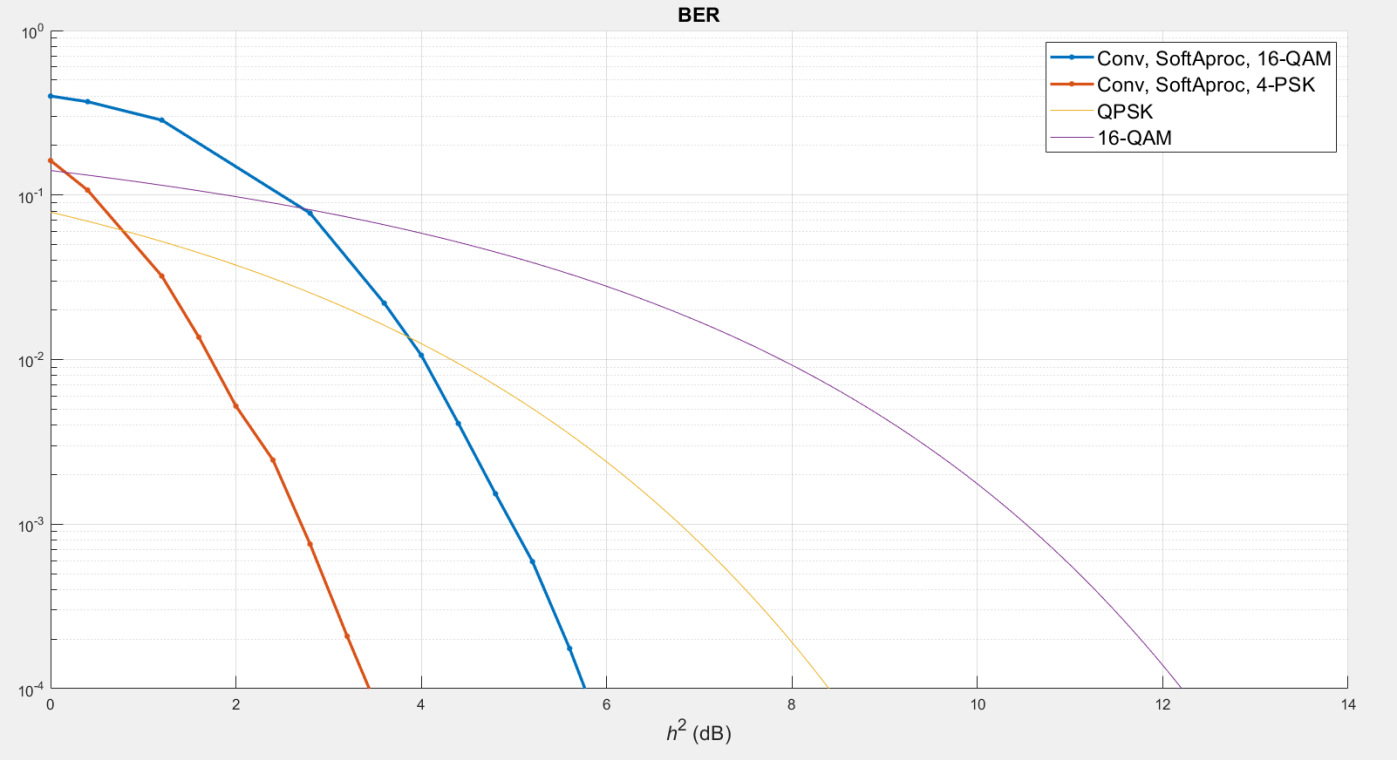


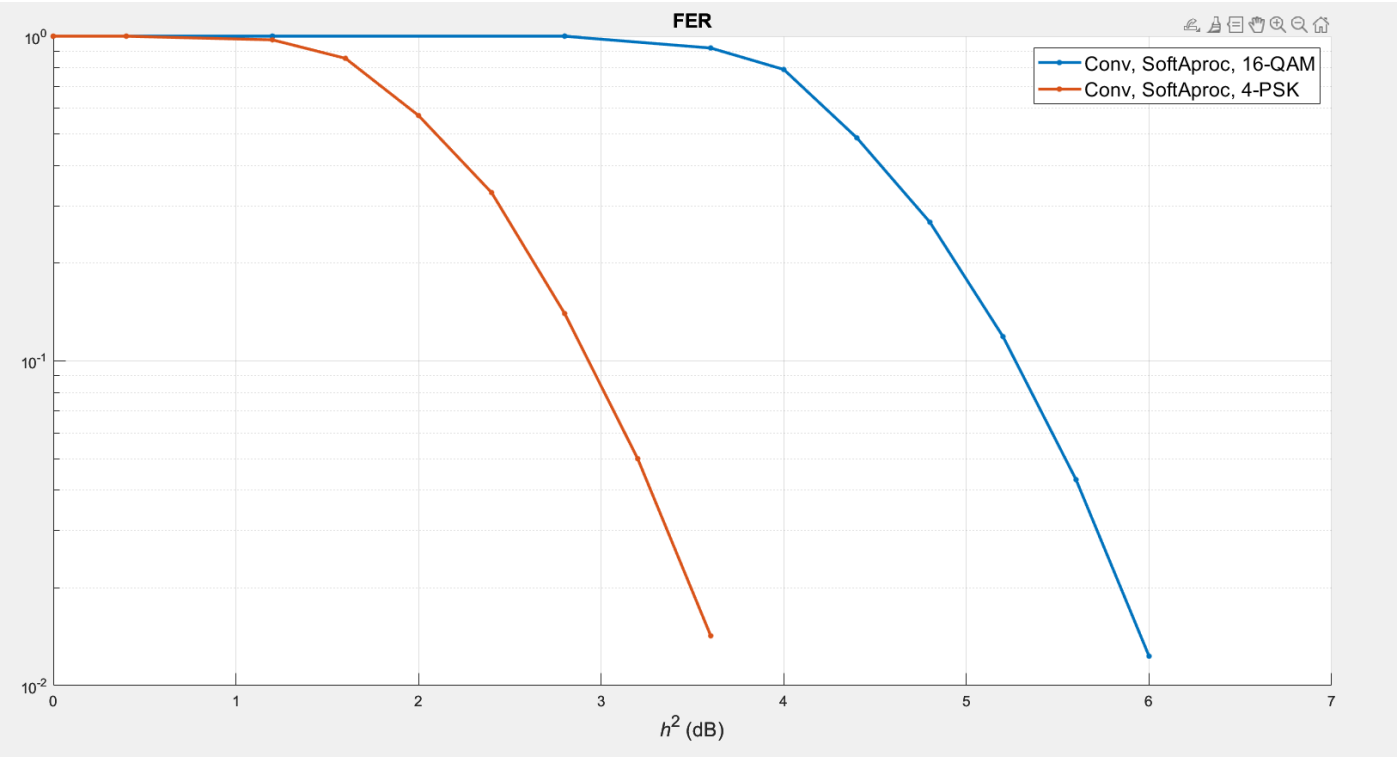
1.2) Мягкие Точные Решения Демодулятора, сверточный код для созвездий QAM-16 и PSK-4.



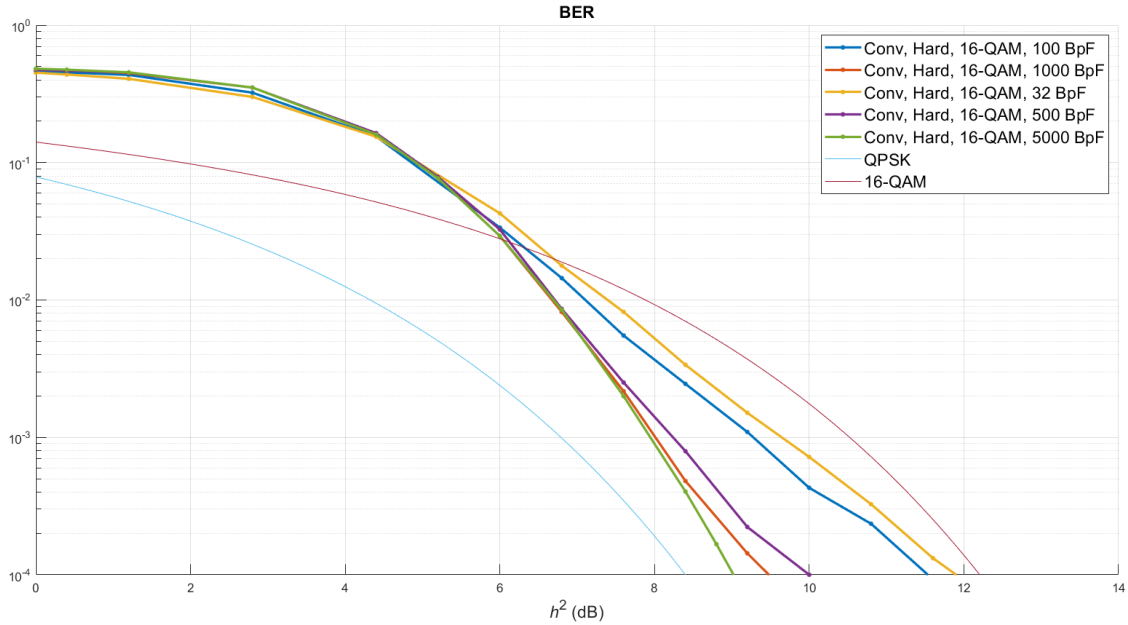


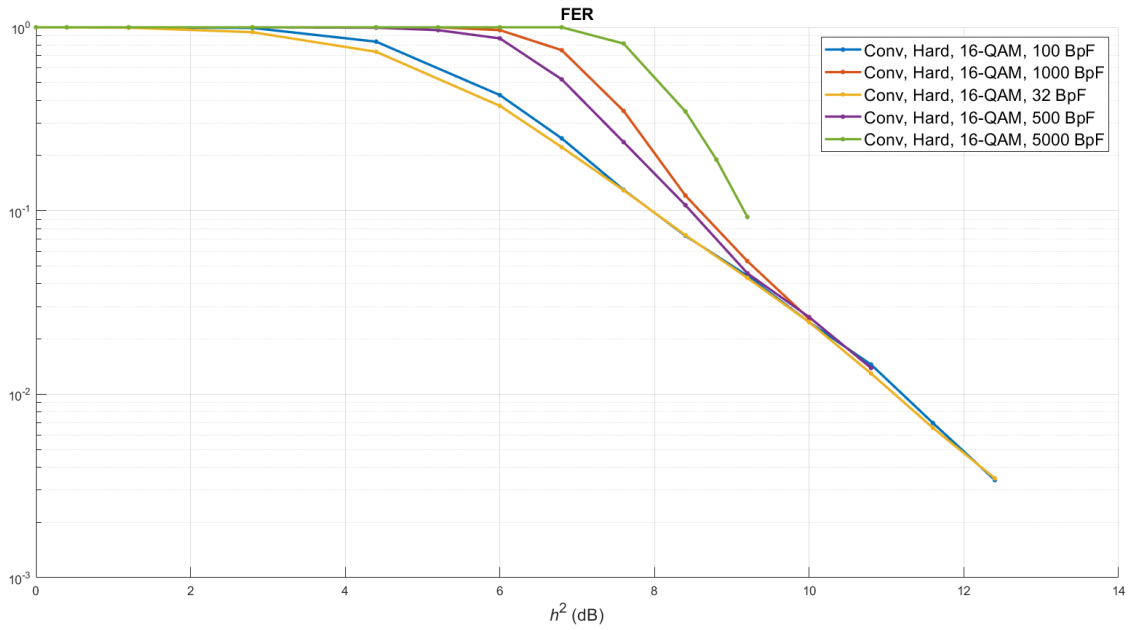
1.3) Мягкие Аппроксимированные Решения Демодулятора, сверточный код для созвездий QAM-16 и PSK-4.



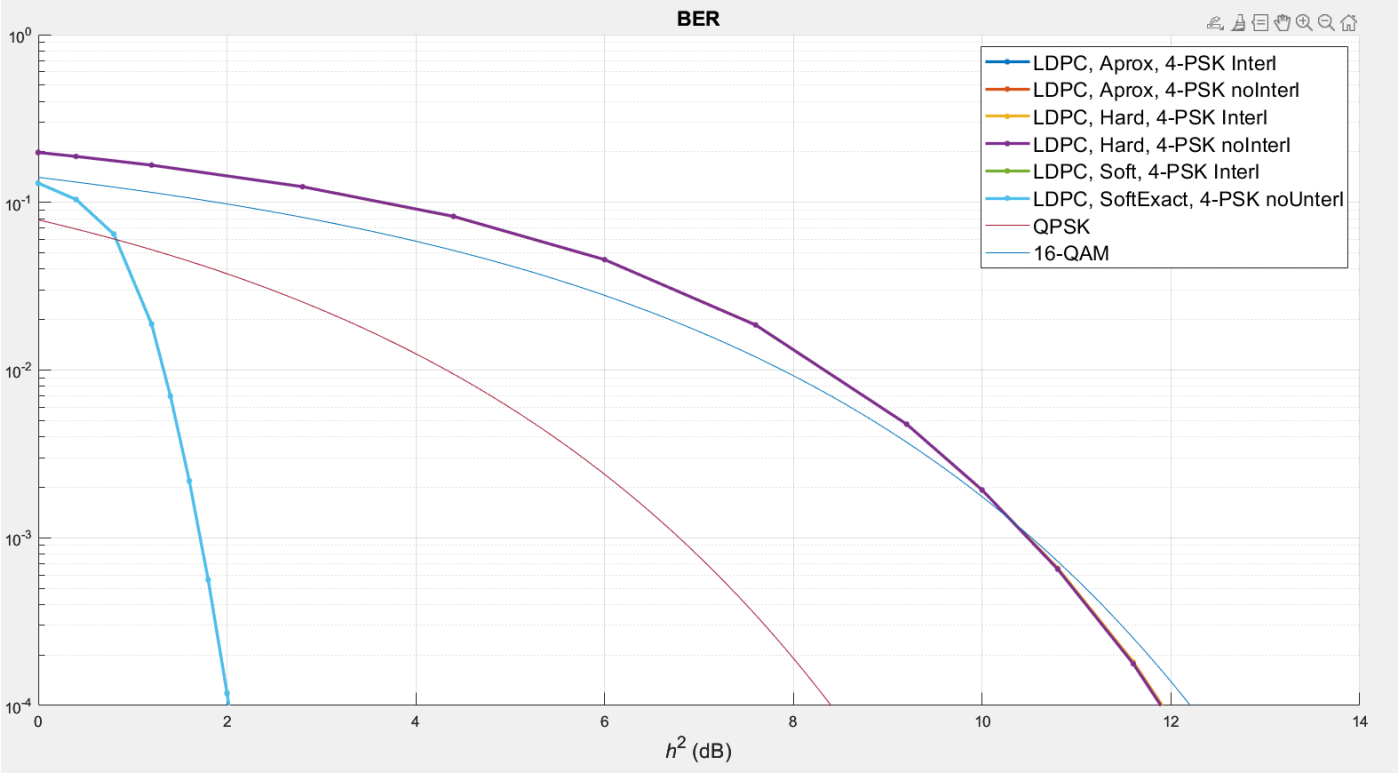


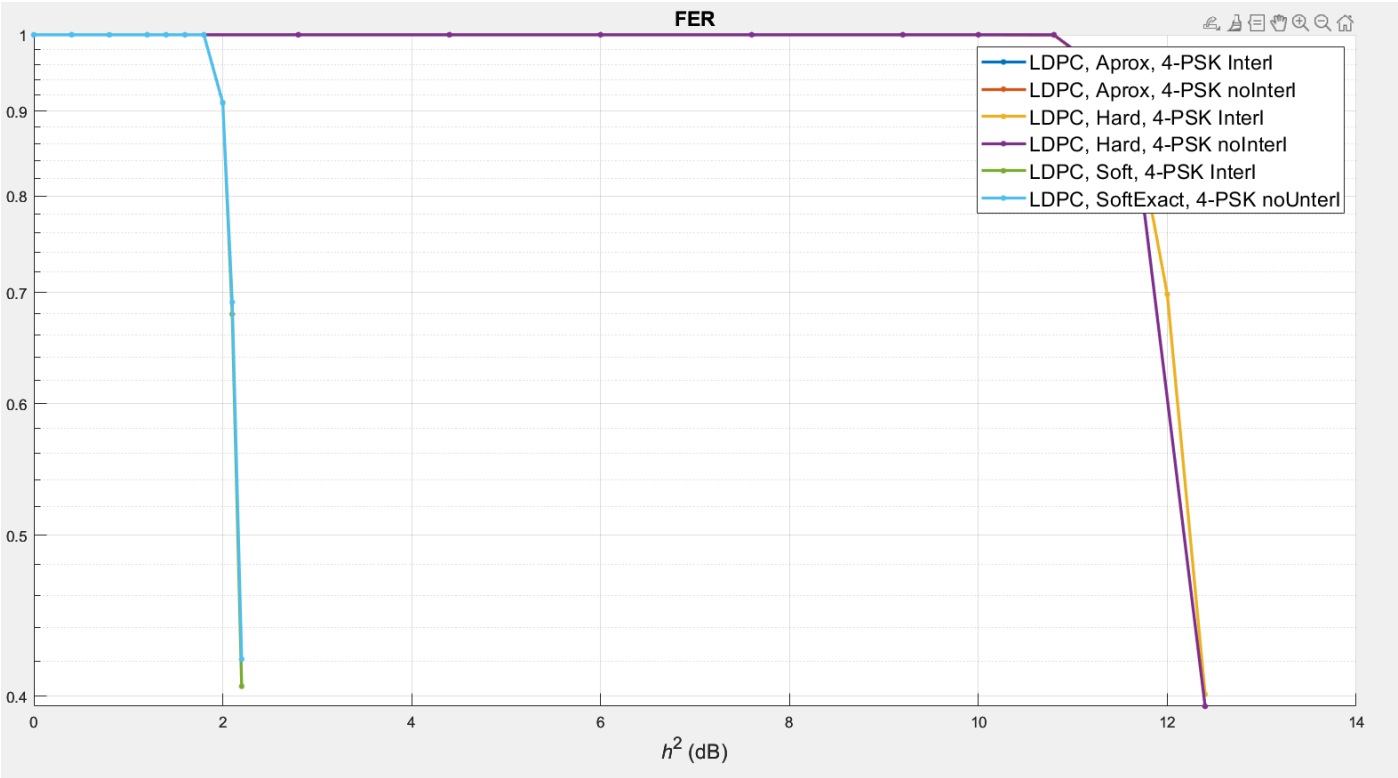
2) Зависимость BER и FER от длины информационного блока



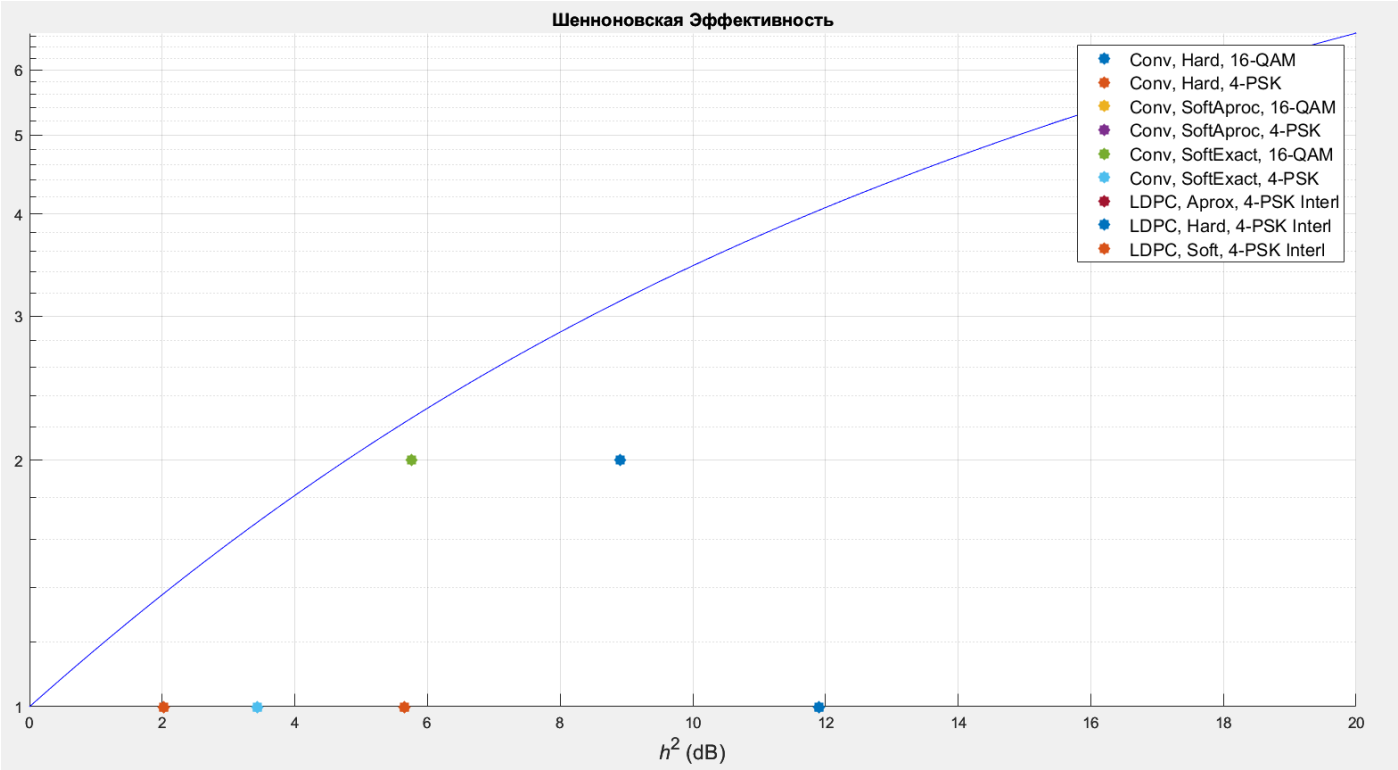


3) LDPC кодер с использованием и без использования перемежителя для разных типов решений, .


BER
LDPC, Aprox, 4-PSK Interl
LDPC, Aprox, 4-PSK noInterl
LDPC, Hard, 4-PSK Interl
LDPC, Hard, 4-PSK noInterl
LDPC, Soft, 4-PSK Interl
LDPC, SoftExact, 4-PSK noUnterl
QPSK
16-QAM
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0
2
4
6
8
10
12
14
h^2 (dB)


FER
LDPC, Aprox, 4-PSK Interl
LDPC, Aprox, 4-PSK noInterl
LDPC, Hard, 4-PSK Interl
LDPC, Hard, 4-PSK noInterl
LDPC, Soft, 4-PSK Interl
LDPC, SoftExact, 4-PSK noUnterl
1
0.9
0.8
0.7
0.6
0.5
0.4
0
2
4
6
8
10
12
14
h^2 (dB)

4)


Шенноновская Эффективность
Conv, Hard, 16-QAM
Conv, Hard, 4-PSK
Conv, SoftAproc, 16-QAM
Conv, SoftAproc, 4-PSK
Conv, SoftExact, 16-QAM
Conv, SoftExact, 4-PSK
LDPC, Aprox, 4-PSK Interl
LDPC, Hard, 4-PSK Interl
LDPC, Soft, 4-PSK Interl
1
2
3
4
5
6
0
2
4
6
8
10
12
14
16
18
20
h^2 (dB)